VIII. Finite element formulations in 3D

1. 3D weak form in mechanics
2. FE approximation
3. Isoparametric concept
4. Master element
5. Discretized weak form: FE force vectors
6. Newton-Raphson iteration
7. Numerical quadrature
8. Alternative weak form descriptions

Weak form in 3D

Deformation $\boldsymbol{x} = \boldsymbol{\varphi}(\boldsymbol{X}) = \boldsymbol{X} + \boldsymbol{u}$

Variation of the deformation $\boldsymbol{\varphi}_\eta = \boldsymbol{\varphi} + \eta\,\delta\boldsymbol{\varphi}$

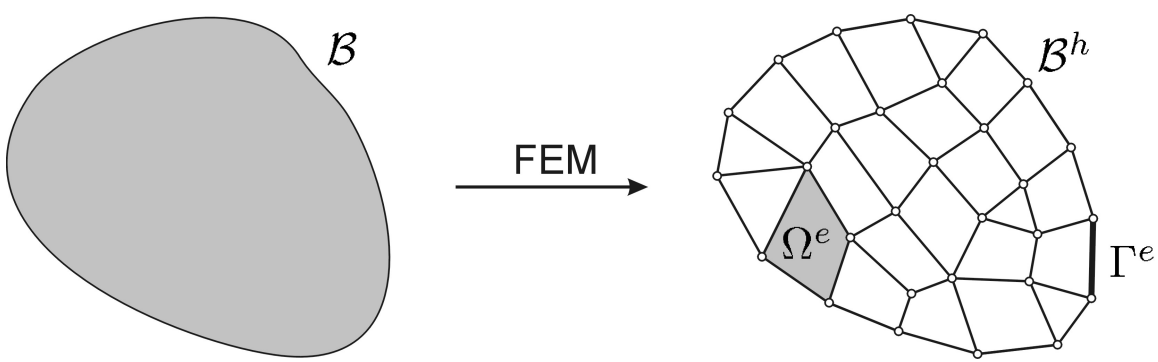
$\delta\boldsymbol{\varphi} = \delta\boldsymbol{u} = \boldsymbol{v} \in \mathcal{V}\,,\quad \mathcal{V} =$ space of kinematically admissible variations

Strong form $\operatorname{div}\boldsymbol{\sigma} + \rho\bar{\boldsymbol{b}} = \boldsymbol{0}$ satisfied at every $\boldsymbol{x} \in \mathcal{B}$

Weak form

$$\boxed{\underbrace{\int_{\mathcal{B}} \operatorname{grad}(\boldsymbol{v}) : \boldsymbol{\sigma}\,\mathrm{d}v}_{\substack{\text{Internal virtual work}\\ G_{\mathrm{int}}}} \underbrace{- \int_{\mathcal{B}} \boldsymbol{v}\cdot\rho\,\bar{\boldsymbol{b}}\,\mathrm{d}v - \int_{\partial_t\mathcal{B}} \boldsymbol{v}\cdot\bar{\boldsymbol{t}}\,\mathrm{d}a}_{\substack{-\text{ External virtual work}\\ -G_{\mathrm{ext}}}} = 0 \qquad \forall\,\boldsymbol{v} \in \mathcal{V}} \qquad \boldsymbol{\sigma} = \boldsymbol{\sigma}(\boldsymbol{F})$$

FE discretization (Lagrangian)



FE approximation

FE approximation for the displacement within an element

$$\boldsymbol{u} \approx \boldsymbol{u}^h = \sum_{A=1}^{n} N_A \, \boldsymbol{u}_A = \mathbf{N} \, \mathbf{u}_e \qquad \mathbf{N} := \left[N_1 \boldsymbol{I} \, , \; N_2 \boldsymbol{I} \, , \; \cdots \; N_n \boldsymbol{I} \, \right] \qquad \mathbf{u}_e := \begin{bmatrix} \boldsymbol{u}_1 \\ \boldsymbol{u}_2 \\ \vdots \\ \boldsymbol{u}_n \end{bmatrix}$$

$n =$ number of nodes per element ($=$ 4 for 2D linear quad elements)

FE shape functions (e.g. bi-linear, 2D quad)

$$N_I(\xi) = (1 + \xi_I \xi)(1 + \eta_I \eta)/4$$

FE approximation for the variation – Galerkin formulation

$$\boldsymbol{v} \approx \boldsymbol{v}^h = \sum_{A=1}^{n} N_A \, \boldsymbol{v}_A = \mathbf{N} \, \mathbf{v}_e$$

FE approximation for the geometry – Isoparameteric concept

$$\boldsymbol{X} \approx \boldsymbol{X}^h = \sum_{A=1}^{n} N_A \, \boldsymbol{X}_A = \mathbf{N} \, \mathbf{X}_e$$

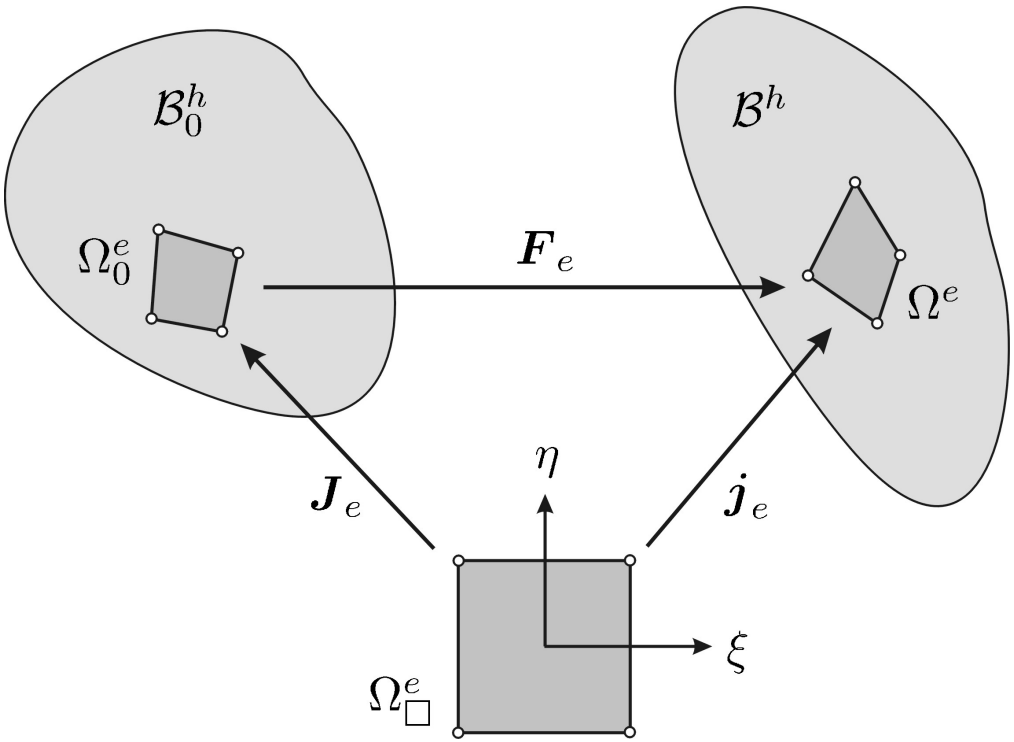
$$\boldsymbol{x} \approx \boldsymbol{x}^h = \sum_{A=1}^{n} N_A \, \boldsymbol{x}_A = \mathbf{N} \, \mathbf{x}_e$$

Summary

$$\boldsymbol{u} \approx \mathbf{N} \, \mathbf{u}_e$$
$$\boldsymbol{v} \approx \mathbf{N} \, \mathbf{v}_e$$
$$\boldsymbol{X} \approx \mathbf{N} \, \mathbf{X}_e$$
$$\boldsymbol{x} \approx \mathbf{N} \, \mathbf{x}_e$$

Master element

Master element



Various mappings

$$\boldsymbol{J}_e = \frac{\partial \boldsymbol{X}}{\partial \boldsymbol{\xi}} \approx \sum_A^n \mathbf{X}_A \otimes \frac{\partial N_A}{\partial \boldsymbol{\xi}}$$

$$\boldsymbol{j}_e = \frac{\partial \boldsymbol{x}}{\partial \boldsymbol{\xi}} \approx \sum_A^n \mathbf{x}_A \otimes \frac{\partial N_A}{\partial \boldsymbol{\xi}}$$

$$\boldsymbol{F}_e = \boldsymbol{j}_e \boldsymbol{J}_e^{-1}$$

Mapping of derivatives

$$\frac{\partial N_A}{\partial \boldsymbol{x}} = \boldsymbol{j}_e^{-T} \frac{\partial N_A}{\partial \boldsymbol{\xi}}$$

$$\frac{\partial N_A}{\partial \boldsymbol{X}} = \boldsymbol{J}_e^{-T} \frac{\partial N_A}{\partial \boldsymbol{\xi}}$$

Discretized weak form

Internal virtual work

$$G_{\text{int}} \approx G^h_{\text{int}} = \sum_{e=1}^{n_{\text{vel}}} G^e_{\text{int}}$$

n_{vel} = number of volume elements Ω^e in $\mathcal{B}^h$

with $$G^e_{\text{int}} = \int_{\Omega^e} \text{grad}\,(\boldsymbol{v}) : \boldsymbol{\sigma}\,\mathrm{d}v$$

$$\mathbf{B}_e := \Big[\mathbf{B}_1\,,\quad \mathbf{B}_2\,,\quad \cdots\quad \mathbf{B}_n\Big]$$

$$G^e_{\text{int}} = \mathbf{v}_e^T\,\mathbf{f}^e_{\text{int}}$$

$$\mathbf{f}^e_{\text{int}} = \int_{\Omega^e} \mathbf{B}_e^T \boldsymbol{\sigma}_{\text{v}}\,\mathrm{d}v$$

$$\mathbf{B}_A = \begin{bmatrix} N_{A,x_1} & 0 & 0 \\ 0 & N_{A,x_2} & 0 \\ 0 & 0 & N_{A,x_3} \\ 0 & N_{A,x_3} & N_{A,x_2} \\ N_{A,x_3} & 0 & N_{A,x_1} \\ N_{A,x_2} & N_{A,x_1} & 0 \end{bmatrix}$$

External virtual work

$$G_{\text{ext}} \approx G^h_{\text{ext}} = \sum_{e=1}^{n_{\text{el}}} G^e_{\text{ext}}\,, \qquad G^e_{\text{ext}} = \mathbf{v}_e^T\,\mathbf{f}^e_{\text{ext}}\,, \qquad \mathbf{f}^e_{\text{ext}} = \int_{\Omega^e} \mathbf{N}^T\,\rho\,\bar{\boldsymbol{b}}\,\mathrm{d}v + \int_{\Gamma^e} \mathbf{N}^T\,\bar{\boldsymbol{t}}\,\mathrm{d}a$$

Assembly of force vectors

$$\mathbf{f}_{\text{int}} = \sum_{e=1}^{n_{\text{el}}} \mathbf{f}^e_{\text{int}}\,, \qquad \mathbf{f}_{\text{ext}} = \sum_{e=1}^{n_{\text{el}}} \mathbf{f}^e_{\text{ext}}$$

Discretized weak form

$$\mathbf{v}^T\Big[\mathbf{f}_{\text{int}} - \mathbf{f}_{\text{ext}}\Big] = 0\,, \qquad \forall\,\mathbf{v} \in \mathcal{V}^h$$

Solution: Newton-Raphson

Discretized weak form $\mathbf{v}^T \left[\mathbf{f}_{\text{int}} - \mathbf{f}_{\text{ext}}\right] = 0 \ , \quad \forall\, \mathbf{v} \in \mathcal{V}^h$

Nonlinear equation $\boxed{\mathbf{f}(\mathbf{u}) = \mathbf{f}_{\text{int}} - \mathbf{f}_{\text{ext}} = \mathbf{0}}$ (typically only $\mathbf{f}_{\text{int}} = \mathbf{f}_{\text{int}}(\mathbf{u})$)

Taylor expansion

$$\mathbf{u}_{k+1} = \mathbf{u}_k + \Delta\mathbf{u}_{k+1}$$

$$\mathbf{f}(\mathbf{u}_{k+1}) = \mathbf{f}(\mathbf{u}_k) + \left.\frac{\partial \mathbf{f}}{\partial \mathbf{u}}\right|_{\mathbf{u}_k} \Delta\mathbf{u}_{k+1} + \ldots = \mathbf{0} \qquad \Rightarrow \qquad \left.\frac{\partial \mathbf{f}}{\partial \mathbf{u}}\right|_{\mathbf{u}_k} \Delta\mathbf{u}_{k+1} = -\mathbf{f}(\mathbf{u}_k)$$

Newton Raphson iteration

1. provide starting guess $\mathbf{u}_0$

iterate $k \to k+1$:

2. solve $\boxed{\mathbf{K}(\mathbf{u}_k)\,\Delta\mathbf{u}_{k+1} = -\mathbf{f}(\mathbf{u}_k)}$ for $\Delta\mathbf{u}_{k+1}$
3. update $\mathbf{u}_{k+1} = \mathbf{u}_k + \Delta\mathbf{u}_{k+1}$
4. check for convergence monitoring $\Delta\mathbf{u}_{k+1}$ or $\mathbf{f}(\mathbf{u}_{k+1})$

Tangent matrix

$$\mathbf{K} := \frac{\partial \mathbf{f}}{\partial \mathbf{u}} = \mathbf{K}_{\text{int}} + \mathbf{K}_{\text{ext}}$$

Assembly

$$\mathbf{K}_{\text{int}} = \sum_{e=1}^{n_{\text{el}}} \mathbf{K}_{\text{int}}^e$$

$$\mathbf{K}_{\text{int}}^e = \frac{\partial \mathbf{f}_{\text{int}}^e}{\partial \mathbf{u}_e}$$

Note the essential BC need to be included, otherwise $\mathbf{K}$ is singular

if $\mathbf{f}(\mathbf{u}) = 0$ is linear, the iteration converges in one step

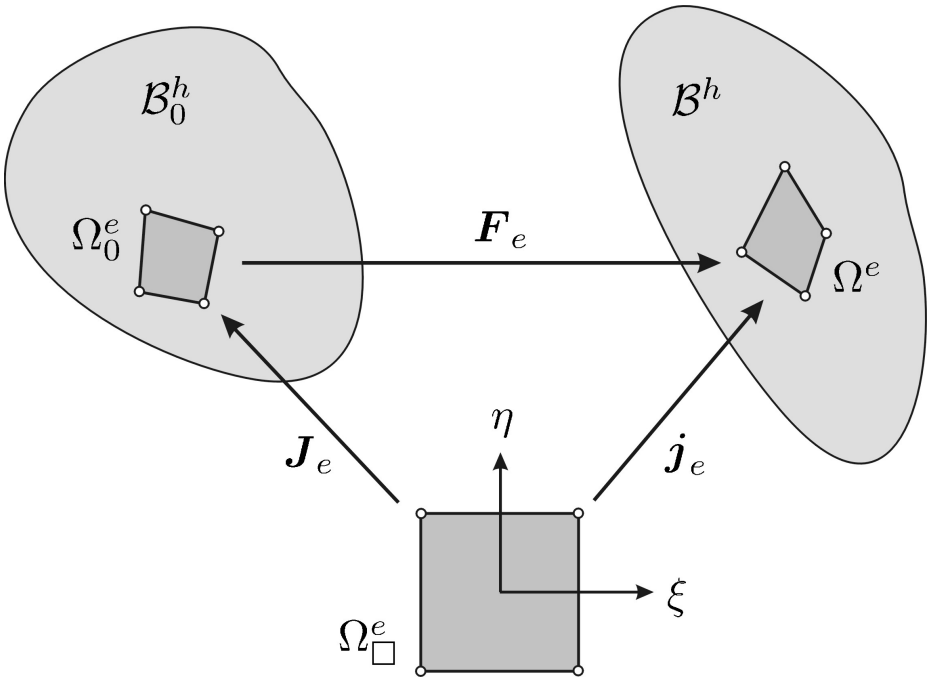
Numerical quadrature

Quadrature formula

$$\int_{\Omega^e_\square} \boldsymbol{g}(\boldsymbol{\xi})\,\mathrm{d}\square \approx \sum_{qp=1}^{n_{\mathrm{qp}}} \boldsymbol{g}(\boldsymbol{\xi}_{qp})\,w_{qp}$$

Gaussian quadrature, for example with

$$n_{\mathrm{qp}} = 2, \quad \xi_{1/2} = \pm 1/\sqrt{3}, \quad w_{1/2} = 1$$



Example

$$\mathbf{f}^e_{\mathrm{int}} = \int_{\Omega^e} \mathbf{B}_e^T \boldsymbol{\sigma}\,\mathrm{d}v = \int_{\Omega^e_0} \mathbf{B}_e^T \boldsymbol{\tau}\,\mathrm{d}V$$

$$= \int_{\Omega^e_\square} \mathbf{B}_e^T \boldsymbol{\sigma}\,\det \boldsymbol{j}_e\,\mathrm{d}\square = \int_{\Omega^e_\square} \mathbf{B}_e^T \boldsymbol{\tau}\,\det \boldsymbol{J}_e\,\mathrm{d}\square$$

$$\approx \sum_{qp=1}^{n_{\mathrm{qp}}} \mathbf{B}_e^T(\boldsymbol{\xi}_{qp})\,\boldsymbol{\sigma}(\boldsymbol{\xi}_{qp})\,\det \boldsymbol{j}_e(\boldsymbol{\xi}_{qp})\,w_{qp}$$

$$\begin{aligned}
\mathrm{d}v &= J\,\mathrm{d}V\,, \quad J = \det \boldsymbol{F} \\
\mathrm{d}v &= \det \boldsymbol{j}_e\,\mathrm{d}\square \\
\mathrm{d}V &= \det \boldsymbol{J}_e\,\mathrm{d}\square \\
\boldsymbol{\tau} &= J\,\boldsymbol{\sigma}
\end{aligned}$$

Discretized weak form: Alternatives

Spatial description (integrated over the spatial domain)

$$\int_{\mathcal{B}} \operatorname{grad}(\delta\boldsymbol{\varphi}) : \boldsymbol{\sigma}\,\mathrm{d}v = \int_{\mathcal{B}} \delta\boldsymbol{\varphi}\cdot\rho\,\bar{\boldsymbol{b}}\,\mathrm{d}v + \int_{\partial_t\mathcal{B}} \delta\boldsymbol{\varphi}\cdot\bar{\boldsymbol{t}}\,\mathrm{d}a\ , \quad \forall\ \delta\boldsymbol{\varphi}\in\mathcal{V}$$

Reformulation of integrals

$$\int_{\mathcal{B}} \delta\boldsymbol{\varphi}\cdot\rho\,\bar{\boldsymbol{b}}\,\mathrm{d}v + \int_{\partial_t\mathcal{B}} \delta\boldsymbol{\varphi}\cdot\bar{\boldsymbol{t}}\,\mathrm{d}a = \int_{\mathcal{B}_0} \delta\boldsymbol{\varphi}\cdot\rho_0\bar{\boldsymbol{b}}\,\mathrm{d}V + \int_{\partial_t\mathcal{B}_0} \delta\boldsymbol{\varphi}\cdot\bar{\boldsymbol{T}}\,\mathrm{d}A$$

$$\begin{aligned}\int_{\mathcal{B}} \operatorname{grad}(\delta\boldsymbol{\varphi}) : \boldsymbol{\sigma}\,\mathrm{d}v &= \int_{\mathcal{B}_0} \operatorname{grad}(\delta\boldsymbol{\varphi}) : \boldsymbol{\tau}\,\mathrm{d}V \\ &= \int_{\mathcal{B}_0} \operatorname{Grad}(\delta\boldsymbol{\varphi}) : \boldsymbol{P}\,\mathrm{d}V = \int_{\mathcal{B}_0} \delta\boldsymbol{F} : \boldsymbol{P}\,\mathrm{d}V = \int_{\mathcal{B}_0} \delta\boldsymbol{E} : \boldsymbol{S}\,\mathrm{d}V\end{aligned}$$

Material description of the weak form

$$\int_{\mathcal{B}_0} \delta\boldsymbol{F} : \boldsymbol{P}\,\mathrm{d}V = \int_{\mathcal{B}_0} \delta\boldsymbol{E} : \boldsymbol{S}\,\mathrm{d}V = \int_{\mathcal{B}_0} \delta\boldsymbol{\varphi}\cdot\rho_0\bar{\boldsymbol{b}}\,\mathrm{d}V + \int_{\partial_t\mathcal{B}_0} \delta\boldsymbol{\varphi}\cdot\bar{\boldsymbol{T}}\,\mathrm{d}A\ , \quad \forall\ \delta\boldsymbol{\varphi}\in\mathcal{V}$$

Corresponding internal FE force vector

$$\mathbf{f}^e_{\mathrm{int}} = \int_{\Omega^e_0} \mathbf{B}^T_{\mathrm{TL}}\,\boldsymbol{S}_{\mathrm{v}}\,\mathrm{d}V$$

Discretized weak form

$$\mathbf{v}^T\left[\mathbf{f}_{\mathrm{int}} - \mathbf{f}_{\mathrm{ext}}\right] = 0\ , \quad \forall\,\mathbf{v}\in\mathcal{V}^h$$

Internal FE force vector: alternatives

Based on the spatial gradient (updated Lagrangian)

$$G^e_{\text{int}} = \int_{\Omega^e} \operatorname{grad}(\delta\boldsymbol{\varphi}) : \boldsymbol{\sigma}\,\mathrm{d}v = \int_{\Omega^e_0} \operatorname{grad}(\delta\boldsymbol{\varphi}) : \boldsymbol{\tau}\,\mathrm{d}V = \mathbf{v}_e^T\,\mathbf{f}^e_{\text{int}}$$

$$\mathbf{f}^e_{\text{int}} = \int_{\Omega^e} \mathbf{B}^T_{\text{UL}}\boldsymbol{\sigma}_{\text{v}}\,\mathrm{d}v \int_{\Omega^e_0} \mathbf{B}^T_{\text{UL}}\boldsymbol{\tau}_{\text{v}}\,\mathrm{d}V$$

$$\mathbf{B}_{\text{UL}} := \left[\mathbf{B}^1_{\text{UL}}\,, \quad \cdots \quad \mathbf{B}^n_{\text{UL}}\right]$$

$$\mathbf{B}^A_{\text{UL}} = \begin{bmatrix} N_{A,x_1} & 0 & 0 \\ 0 & N_{A,x_2} & 0 \\ 0 & 0 & N_{A,x_3} \\ 0 & N_{A,x_3} & N_{A,x_2} \\ N_{A,x_3} & 0 & N_{A,x_1} \\ N_{A,x_2} & N_{A,x_1} & 0 \end{bmatrix}$$

Based on the material gradient (total Lagrangian)

$$G^e_{\text{int}} = \int_{\Omega^e_0} \delta\boldsymbol{E} : \boldsymbol{S}\,\mathrm{d}V = \int_{\Omega^e_0} \delta\boldsymbol{F} : \boldsymbol{P}\,\mathrm{d}V = \mathbf{v}_e^T\,\mathbf{f}^e_{\text{int}}$$

$$\mathbf{f}^e_{\text{int}} = \int_{\Omega^e_0} \mathbf{B}^T_{\text{TL}}\,\boldsymbol{S}_{\text{v}}\,\mathrm{d}V = \int_{\Omega^e_0} \mathbf{B}^T_{\text{TLP}}\,\boldsymbol{P}_{\text{v}}\,\mathrm{d}V$$

$$\mathbf{B}^A_{\text{TL}} = \begin{bmatrix} \boldsymbol{\varphi}^T_{,X_1}\,N_{A,X_1} \\ \boldsymbol{\varphi}^T_{,X_2}\,N_{A,X_2} \\ \boldsymbol{\varphi}^T_{,X_3}\,N_{A,X_3} \\ \boldsymbol{\varphi}^T_{,X_2}\,N_{A,X_3} + \boldsymbol{\varphi}^T_{,X_3}\,N_{A,X_2} \\ \boldsymbol{\varphi}^T_{,X_3}\,N_{A,X_1} + \boldsymbol{\varphi}^T_{,X_1}\,N_{A,X_3} \\ \boldsymbol{\varphi}^T_{,X_1}\,N_{A,X_2} + \boldsymbol{\varphi}^T_{,X_2}\,N_{A,X_1} \end{bmatrix}$$

$$\mathbf{B}^A_{\text{TLP}} = \begin{bmatrix} N_{A,X_1} & 0 & 0 \\ 0 & N_{A,X_2} & 0 \\ 0 & 0 & N_{A,X_3} \\ N_{A,X_2} & 0 & 0 \\ 0 & N_{A,X_1} & 0 \\ 0 & N_{A,X_3} & 0 \\ 0 & 0 & N_{A,X_2} \\ 0 & 0 & N_{A,X_1} \\ N_{A,X_3} & 0 & 0 \end{bmatrix}$$